Community detection in geometric graphs: Recent advances until 2026

Raphaël Sala

Affiliation à renseigner
raphael.sala@utoulouse.fr

Résumé TODO

Keywords: Community detection, geometric graphs, random geometric graphs, exact/partial recovery, spectral methods

1 Introduction

General idea Community detection consists to find clusters in a graph where nodes share more connections with nodes of the same cluster than with nodes of other clusters. This large topic is detailled in general by [Fortunato, 2010]. In this report we focus on random geometric graphs (RGG). The specificity to add spatial information allows us to consider local information based on real or abstract proximity. In social networks, two individuals are more susceptible to be friends if they are geographically near or if they have similar terms of interests, age, ... For more concrete examples of RGG can be find for wireless networks from [Gomez et al., 2015], biology [Higham et al., 2008] and in particular an application of detection of communities [Sankararaman et al., 2020] word embedding, physic, ...

Why geometric The main model for community detection is the Stochastic Block Model (SBM) [Holland et al., 1983]. Each node carries a hidden community label and edge probabilities depend on whether two nodes belong to the same community or not. This model comes from the Erdos-Renyi graph, where each node connect with the same probability to others nodes [Erdős et al., 1960]. This model is nowadays well understood, see [Abbe, 2018]. Nonetheless, it suffers to some drawbacks because its unrealistic to think that every nodes can connected to others with the same probability rate. At the same time than Erdos-Renyi, the RGG was introduced by [Gilbert, 1961]. Add geometric information allows us to mimic some emergent behaviour from real network, like sparsity [Del Genio et al., 2011], the "small world" property [Watts and Strogatz, 1998] or even transitivity "friend of a friend is a friend" Ref ?.

Context of the problem RGG and its properties are well studied, see [Penrose, 2003]. When we add community structure, we are in a different setting. We observe the graph, and sometimes also the node locations, while the community labels remain hidden. Naturally, we want to know if we can recover the latent labels, what is the best accuracy we can achieve, and which algorithms can reach it.

TODO Définition et interprétation, récupération à permutation près

- Distinguabilité [Sankararaman and Baccelli, 2018]
- Récupération partielle (weak recovery)
- Récupération presque exacte (almost exact recovery)
- Récupération exacte (exact recovery) [Gaudio et al., 2024a]

Thus, in this report, community detection is not only understood as producing a convenient partition of the observed graph, but rather as recovering latent community labels under a probabilistic geometric model, up to permutation.

What we don't do Precedently we enhance properties RGG share with real networks due to the geometric information. However, we can get for instance transitivity, and surely more, thanks to attribute based models [Li et al., 2018] or hyper-graph [Ruggeri et al., 2023] and even with its geometric variant [Kovács et al., 2025]. We will not consider these models to focus on the geometric aspect. Moreover, we presented problems we are interested about. From them, one is *Distinguability* to not confond with the *Detection Problem* we don't treat. This last one ask generally condition on parameters, defining the RGG, to be distinguable to an Erdos-Renyi graph. Hence, this hasn't link to community. Refer to [Bubeck et al., 2016] [Liu et al., 2022] [Mao et al., 2026]. Finally, at the beginning of the literature, some works were interested to recover the latent position of nodes in the geometric space, and consider it as a Community detection problem, [Valdivia and De Castro, 2019] [Eldan et al., 2022]. This is a different problem than the one we are interested in, and it is more difficult because we want to recover the position (contnue label) of each node rather than community (discrete label).

2 Modèle

- pas de boucle, pas dirige, pas de communautés chevauchantes, etc...
- espace latent (ou pas) [Castro et al., 2020]
- sphère, espace euclidien ?
- régime : dense, logarithmique, sparse
- hard-RGG, soft-RGG [Duchemin and De Castro, 2023]
- motivation : bien que les modèles ne soient pas les mêmes, ils n'ont pas pour but de comprendre tous ceux possible mais qu'à travers chaque classe voir les mécanismes. Exemplse : passer du local au global puis raffinement, les quantités utilisées, ...

Le but est de faire des notaitons claires, concises et générales pour pouvori expliquer tous les modèles ffff sans avoir à se perdre dans la notation.

3 Seuil théorique information

Faire le rapport dans l'ordre chronologique je pense

— [Galhotra et al., 2018] (Geometric Block Model, GBM)
— [Sankararaman and Baccelli, 2018] & [Abbe et al., 2021] (Geometric SBM)
— [Chien et al., 2020]
— [Galhotra et al., 2020] (GBM)
— [Galhotra et al., 2023] (GBM)
— [Avrachenkov et al., 2024] (Geometric Kernel Block Model, GKBM)
— [Gaudio et al., 2024b] (Geometric SBM)
— [Gaudio et al., 2024a] (Geometric Hidden Community Model, GHCM)
— [Gaudio and Guan, 2025] (GHCM)
— [Gaudio and Jin, 2026] (GHCM)

3.1 Geometric Block Model (GBM)

[Galhotra et al., 2018] [Galhotra et al., 2020] [Galhotra et al., 2023] [Chien et al., 2020]

3.2 Geometric Stochastic Block Model (GSBM)

[Sankararaman and Baccelli, 2018] [Abbe et al., 2021]

3.3 (Geometric Kernel Block Model, GKBM)

[Avrachenkov et al., 2024] [Allem et al., 2025]

3.4 Geometric Hidden Community Model (GHCM)

[Gaudio et al., 2024b] [Gaudio et al., 2024a] [Gaudio and Guan, 2025] [Gaudio and Jin, 2026]

4 Spectral

(A confirmer) [Chen et al., 2016]

(papier SBM mais théorie intéressante) [Bordenave et al., 2015]

Cadre spectral intéressant pour certains espaces [Méliot, 2019]

Méthode spectrale + Robustesse (la perturbation est géométrique) [Péché and Perchet, 2020]

Méthode spectrale sur Soft Geometric Block Model (SGBM) (Generalisation de Geomemetric SBM) [Avrachenkov et al., 2021]

5 Ouvertures

- Considérer des modèles hyperboliques, pas seulement labels & distance-dependant, car ceux-ci captent mieux les propriétés habituelles des réseaux
  - Clustering [Krioukov, 2016]
  - Degre inhomogène [Barabási and Albert, 1999]
  - "Communities with specific mixing patterns" (mesoscale) [Toivonen et al., 2006]

  Communautés apparaissent dans un espace hyperbolique latent [Bruno et al., 2019] Réponse 6ans plus tard : pas toujours ! + definition d'un modele pour "régler" ça [Guarino et al., 2025]

  [Li et al., 2018]

Bibliographie

Emmanuel Abbe. Community detection and stochastic block models : recent developments. *Journal of Machine Learning Research*, 18(177) :1–86, 2018.

Emmanuel Abbe, Francois Baccelli, and Abishek Sankararaman. Community detection on euclidean random graphs. *Information and Inference : A Journal of the IMA*, 10(1) :109–160, 2021.

Luiz Emilio Allem, Konstantin Avrachenkov, Carlos Hoppen, Hariprasad Manjunath, and Lucas Siviero Sibemberg. Multi-community spectral clustering for geometric graphs, 2025. URL `https://arxiv.org/abs/2508.00893`.

Konstantin Avrachenkov, Andrei Bobu, and Maximilien Dreveton. Higher-order spectral clustering for geometric graphs. *Journal of Fourier Analysis and Applications*, 27(2) :22, 2021.

Konstantin Avrachenkov, BR Kumar, and Lasse Leskelä. Community detection on block models with geometric kernels. *arXiv preprint arXiv :2403.02802*, 2024.

Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439) :509–512, October 1999. ISSN 1095-9203. `https://doi.org/10.1126/science.286.5439.509`. URL `http://dx.doi.org/10.1126/science.286.5439.509`.

Charles Bordenave, Marc Lelarge, and Laurent Massoulié. Non-backtracking spectrum of random graphs : community detection and non-regular ramanujan graphs. In *2015 IEEE 56th Annual Symposium on Foundations of Computer Science*, pages 1347–1357. IEEE, 2015.

Matteo Bruno, Sandro Ferreira Sousa, Furkan Gursoy, Matteo Serafino, Francesca V Vianello, Ana Vranić, and Marián Boguñá. Community detection in the hyperbolic space. *arXiv preprint arXiv :1906.09082*, 2019.

Sébastien Bubeck, Jian Ding, Ronen Eldan, and Miklós Z Rácz. Testing for high-dimensional geometry in random graphs. *Random Structures & Algorithms*, 49(3) :503–532, 2016.

Yohann De Castro, Claire Lacour, and Thanh Mai Pham Ngoc. Adaptive estimation of nonparametric geometric graphs, 2020. URL `https://arxiv.org/abs/1708.02107`.

Yuxin Chen, Govinda Kamath, Changho Suh, and David Tse. Community recovery in graphs with locality. In *International conference on machine learning*, pages 689–698. PMLR, 2016.

Eli Chien, Antonia Tulino, and Jaime Llorca. Active learning in the geometric block model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 3641–3648, 2020.

Charo I. Del Genio, Thilo Gross, and Kevin E. Bassler. All scale-free networks are sparse. *Physical Review Letters*, 107(17), October 2011. ISSN 1079-7114. `https://doi.org/10.1103/physrevlett.107.178701`. URL `http://dx.doi.org/10.1103/PhysRevLett.107.178701`.

Quentin Duchemin and Yohann De Castro. Random geometric graph : Some recent developments and perspectives. *High Dimensional Probability IX : The Ethereal Volume*, pages 347–392, 2023.

Ronen Eldan, Dan Mikulincer, and Hester Pieters. Community detection and percolation of information in a geometric setting. *Combinatorics, Probability and Computing*, 31(6) :1048–1069, 2022.

Paul Erdős, Alfréd Rényi, et al. On the evolution of random graphs. *Publications of the*, 1960.

Santo Fortunato. Community detection in graphs. *Physics reports*, 486(3-5) : 75–174, 2010.

Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. The geometric block model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.

Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. Connectivity in random annulus graphs and the geometric block model, 2020. URL `https://arxiv.org/abs/1804.05013`.

Sainyam Galhotra, Arya Mazumdar, Soumyabrata Pal, and Barna Saha. Community recovery in the geometric block model, 2023. URL `https://arxiv.org/abs/2206.11303`.

Julia Gaudio and Charlie K. Guan. Sharp exact recovery threshold for two-community euclidean random graphs, 2025. URL `https://arxiv.org/abs/2501.14830`.

Julia Gaudio and Andrew Jin. Exact recovery in the geometric hidden community model. *arXiv preprint arXiv :2601.17591*, 2026.

Julia Gaudio, Charlie Guan, Xiaochun Niu, and Ermin Wei. Exact label recovery in euclidean random graphs. *arXiv preprint arXiv :2407.11163*, 2024a.

Julia Gaudio, Xiaochun Niu, and Ermin Wei. Exact community recovery in the geometric sbm, 2024b. URL `https://arxiv.org/abs/2307.11196`.

Edward N Gilbert. Random plane networks. *Journal of the society for industrial and applied mathematics*, 9(4) :533–543, 1961.

Jean-Sébastien Gomez, Aurélien Vasseur, Anaïs Vergne, Philippe Martins, Laurent Decreusefond, and Wei Chen. A case study on regularity in cellular network deployment. *IEEE wireless communications letters*, 4(4) :421–424, 2015.

Stefano Guarino, Enrico Mastrostefano, and Davide Torre. Random hyperbolic graphs with arbitrary mesoscale structures. *Physical Review E*, 112(5) :054310, 2025.

Desmond J Higham, Marija Rašajski, and Nataša Pržulj. Fitting a geometric graph to a protein–protein interaction network. *Bioinformatics*, 24(8) :1093–1099, 2008.

Paul W Holland, Kathryn Blackmond Laskey, and Samuel Leinhardt. Stochastic blockmodels : First steps. *Social networks*, 5(2) :109–137, 1983.

Bianka Kovács, Barnabás Benedek, and Gergely Palla. Community detection in hypergraphs through hyperedge percolation. *Scientific Reports*, 15(1), October 2025. ISSN 2045-2322. `https://doi.org/10.1038/s41598-025-19974-9`. URL `http://dx.doi.org/10.1038/s41598-025-19974-9`.

Dmitri Krioukov. Clustering implies geometry in networks. *Physical Review Letters*, 116(20), May 2016. ISSN 1079-7114. `https://doi.org/10.1103/physrevlett.116.208302`. URL `http://dx.doi.org/10.1103/PhysRevLett.116.208302`.

Ye Li, Chaofeng Sha, Xin Huang, and Yanchun Zhang. Community detection in attributed graphs : An embedding approach. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), Apr. 2018.

Siqi Liu, Sidhanth Mohanty, Tselil Schramm, and Elizabeth Yang. Testing thresholds for high-dimensional sparse random geometric graphs. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, pages 672–677, 2022.

Cheng Mao, Yihong Wu, and Jiaming Xu. Random geometric graphs with smooth kernels : sharp detection threshold and a spectral conjecture, 2026. URL `https://arxiv.org/abs/2602.14998`.

Pierre-Loïc Méliot. Asymptotic representation theory and the spectrum of a random geometric graph on a compact lie group. 2019.

Sandrine Péché and Vianney Perchet. Robustness of community detection to random geometric perturbations. *Advances in Neural Information Processing Systems*, 33 :17827–17837, 2020.

Mathew Penrose. *Random geometric graphs*, volume 5. OUP Oxford, 2003.

Nicolò Ruggeri, Martina Contisciani, Federico Battiston, and Caterina De Bacco. Community detection in large hypergraphs. *Science Advances*, 9(28), July 2023. ISSN 2375-2548. `https://doi.org/10.1126/sciadv.adg9159`. URL `http://dx.doi.org/10.1126/sciadv.adg9159`.

Abishek Sankararaman and François Baccelli. Community detection on euclidean random graphs. In *Proceedings of the Twenty-Ninth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 2181–2200. SIAM, 2018.

Abishek Sankararaman, Haris Vikalo, and François Baccelli. Comhapdet : a spatial community detection algorithm for haplotype assembly. *BMC genomics*, 21(Suppl 9) :586, 2020.

Riitta Toivonen, Jukka-Pekka Onnela, Jari Saramäki, Jörkki Hyvönen, and Kimmo Kaski. A model for social networks. *Physica A : Statistical Mechanics and its Applications*, 371(2) :851–860, November 2006. ISSN 0378-4371. `https://doi.org/10.1016/j.physa.2006.03.050`. URL `http://dx.doi.org/10.1016/j.physa.2006.03.050`.

Ernesto Araya Valdivia and Yohann De Castro. Latent distance estimation for random geometric graphs. In *NeurIPS*, pages 8721–8731, 2019.

Duncan J Watts and Steven H Strogatz. Collective dynamics of 'small-world'networks. *nature*, 393(6684) :440–442, 1998.